C3: the self-reducibility ceiling on every heavy-tail mechanism

(scoring table) **Lead.** C3 of `research/notes/ecdlp-general/RESEARCH_PRIME_FIELD_LEADS_20261031.md`, thresholds in `../PLAN.md`. **Kind.** Derivation and scoring; no experiment, no counted measurement in this directory. **Decision.** writeup (the plan names no threshold for C3; the null prediction "all mechanisms below the ceiling" is what the table shows).

This file was written by the synthesis step (2026-11-01) from the two tables the lead agent left (`ceiling_table.csv`, `mechanism_scoring.csv`) and the graph it plotted from them (`ceiling_graph.svg`, script `plot_ceiling.py`, recorded values only). The agent's run was cut off by host load before it wrote this report; no number below is new.

1. The ceiling

Randomising a target is one group operation, R ↦ R + rP, and the discrete logarithm is random self-reducible. If a mechanism is "easy" on a class of targets of measure f, then

- the **mass gain** on the decomposition phase is at most 1/f: a sieve that re-randomises until the target falls in the class already prices the class at 1/f attempts, so no selection rule can buy more than that factor; and
- the **screen budget** is what §2 of the note leaves for finding such a target: the relation phase is allowed p^{1/2 − 1/m} work per relation before it stops beating rho at the MITM baseline (p^{1/2 + 1/m} total over p^{1/m} relations), so a screen that tests one candidate per O(1) operations may test at most p^{1/2 − 1/m} candidates, i.e. it needs f ≥ p^{−(1/2 − 1/m)}.

For a mechanism to move the exponent it needs its per-attempt cost p^{e/m} to drop below the threshold e < m/2 − 1 (note §2); the table converts that into the $\log_2(1/f)$ the class must supply (`attempt_gain_log2_needed`), the mass requirement (`mass_log2_inv_f_min`, which equals the gain needed since the gain is at most 1/f) and the screen budget (`screen_log2_inv_f_max`).

2. Table (from `ceiling_table.csv`, n = $\log_2$ p from 32 to 512; every row scales linearly in n)

m	baseline	e_base	e needed	$\log_2(1/f)$ needed (mass)	$\log_2(1/f)$ allowed (screen)	window
3	MITM	1.5	< 0.5	n/3	n/6	empty
3	ENUM	2.0	< 0.5	n/2	n/6	empty
4	MITM	2.0	< 1.0	n/4	n/4	one boundary point
4	ENUM	3.0	< 1.0	n/2	n/4	empty

At n = 256: m = 3 needs f ≤ 2^{−85.3} but may screen only 2^{42.7} candidates; m = 4 needs f ≤ 2^{−64} and may screen exactly 2^{64}. So at m = 3 no target-conditioned mechanism with an O(1) screen can reach the rho threshold, whatever its tail, and at m = 4 only a mechanism whose class has measure exactly p^{−1/4} and whose screen costs nothing beyond the test sits on the boundary (rho_gain_per_bit column: 1.0 at that point, 2.0 to 3.0 elsewhere, meaning the tail must supply two to three bits of gain per bit of screening it is allowed).

Graph: `ceiling_graph.svg` (mass requirement against screen budget, m = 3 and m = 4, MITM baseline, against n).

3. Scoring of the catalogue (from `mechanism_scoring.csv`, 18 entries)

Disposition	Mechanisms	Reason recorded
close	M012, M023, M044, M009, M035, M048	locus or class measure is bounded (univariate hypersurface: f ≤ D/p; one solving-degree drop is a constant; a split-node chain of about n/m nodes replaces the m-ary relation; the stated c < 1/2 lies outside the screen budget)
sieve-pricing only	M014, M054, M077, M050, M076, M067	gain ≤ 1/f with f constant or poly-log; M067 is undecided by this ceiling under the R8 normalisation and is left to its own Poisson-tail falsifier
unaffected by the R ceiling	M016, M053, M091	not a target-class mechanism (membership design, matrix columns, a distinguisher feature)
not scored	M085, M080, M029	distinguishers on input pairs, or a screen that is not O(1) (BKZ per candidate)

Six closures are recommended; none of the eighteen is placed above the ceiling. M023's own `revisit_when` clause on the atlas card `LEAD-ICB-multi-target-and-portfolio` is the one card edit the scoring touches (the two windows it needs, $c < 1/2 - 1/m$ and $c \geq 1/m$, are disjoint at $m = 3$ and $m = 4$).

4. Limits

- Derivation only. The ceiling is an inequality on measure; it does not say what any mechanism's actual f is, and a mechanism's own falsifier still decides it where the table says "sieve-pricing only".
- The screen budget assumes an O(1) screen. A screen that is cheaper than one group operation does not exist in the counted unit; one that is dearer only shrinks the window.
- No PDF renderer was available to the lead agent; the synthesis step rendered `REPORT.pdf` from this file with Typst in a scratch virtualenv.

5. Reading

Within the §2 arithmetic, target conditioning cannot carry an index-calculus attack over F_p past rho at $m = 3$, and at $m = 4$ it has exactly one boundary point to stand on. Every heavy-tail mechanism in the catalogue is either already priced by a sieve or bounded by the measure of its own locus.